

DSA 0604

Data Handling and Visualization

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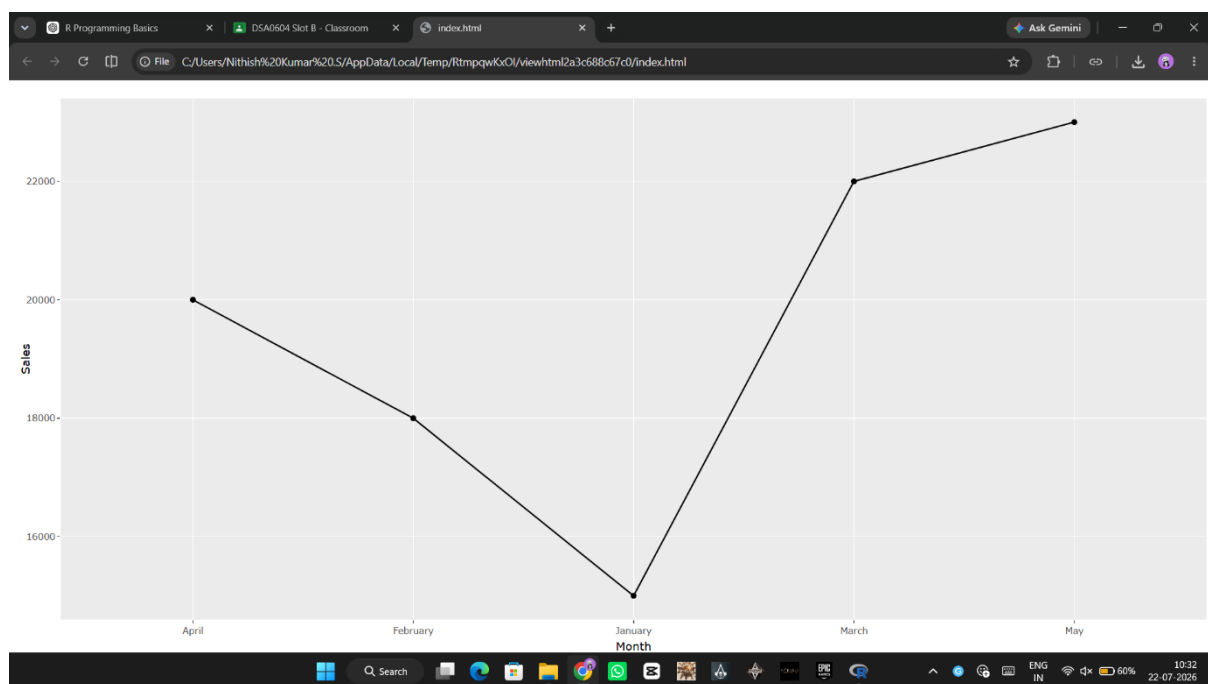
Set 1: Monthly Sales Data

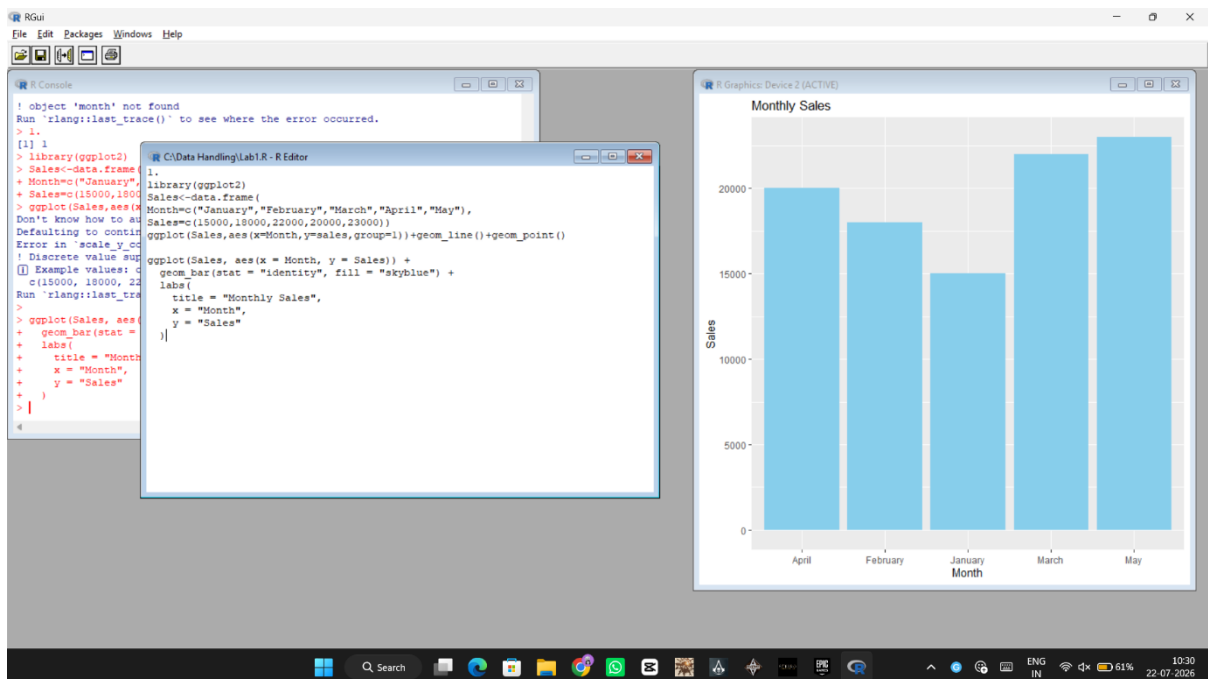
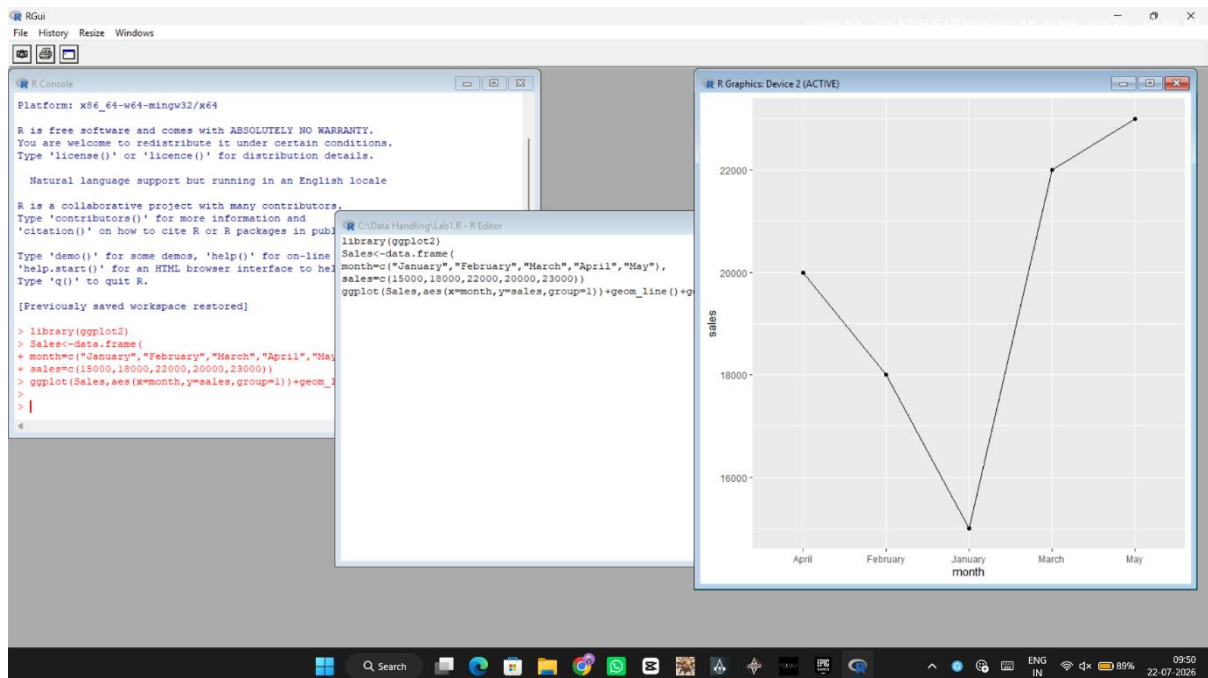
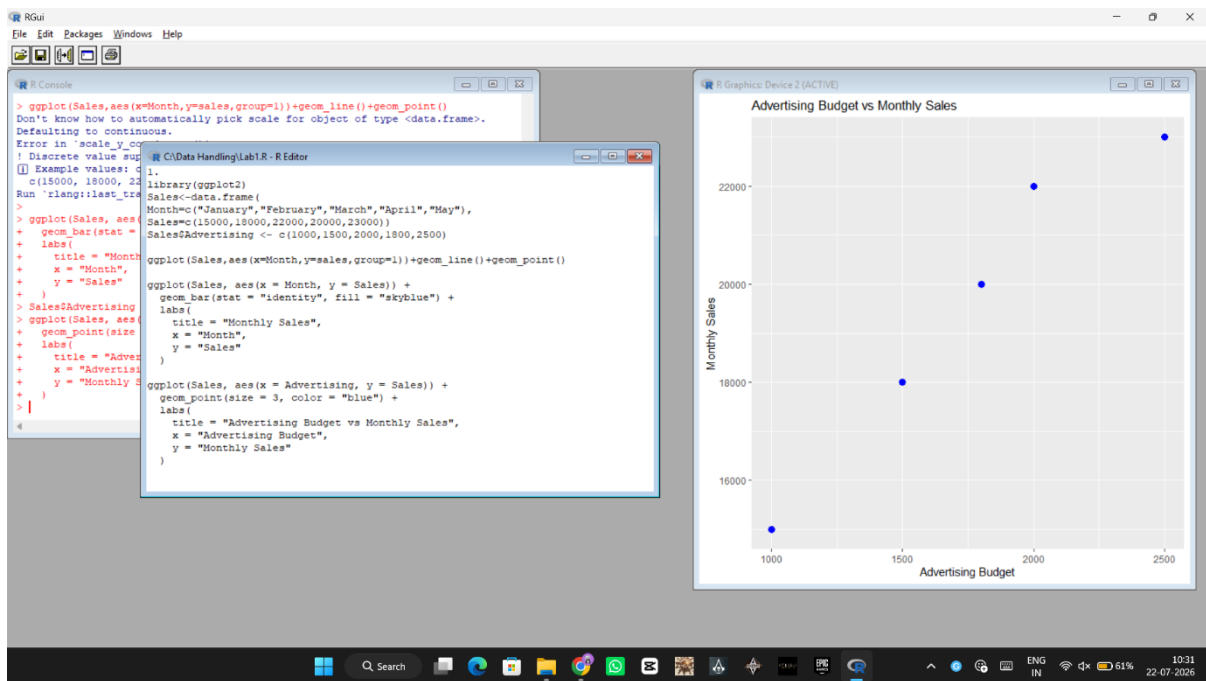
Dataset: Monthly Sales

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.
2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.
3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

Explain the insights drawn from the scatter plot.

4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

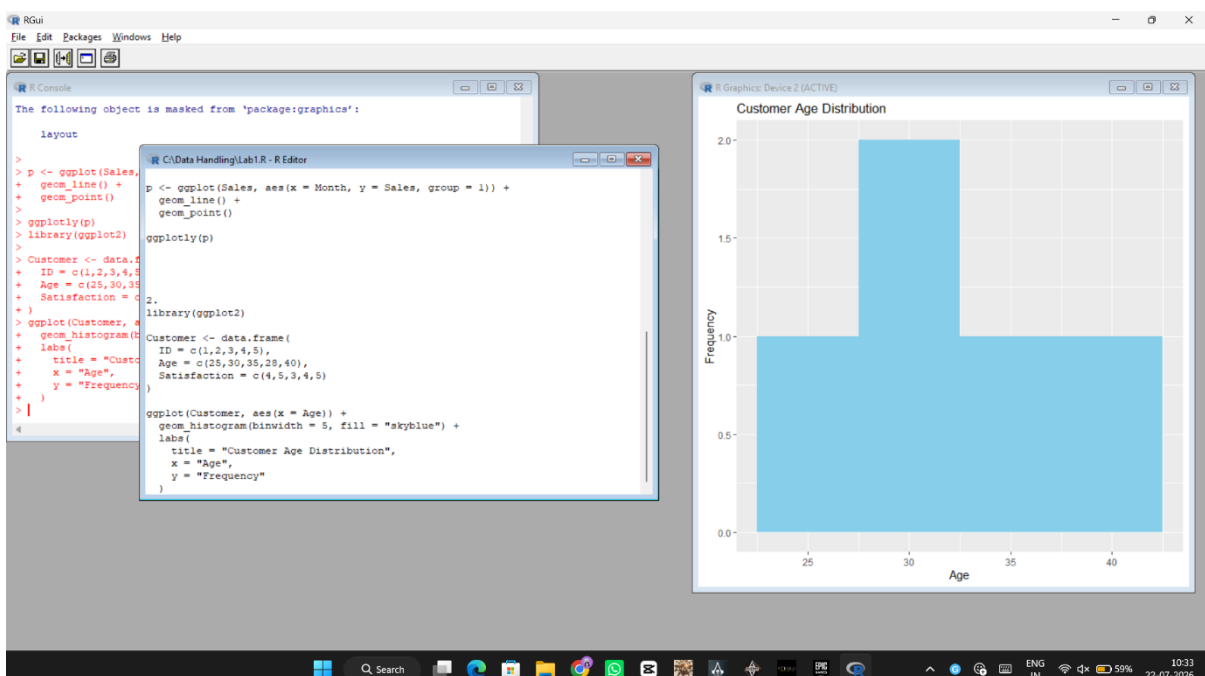
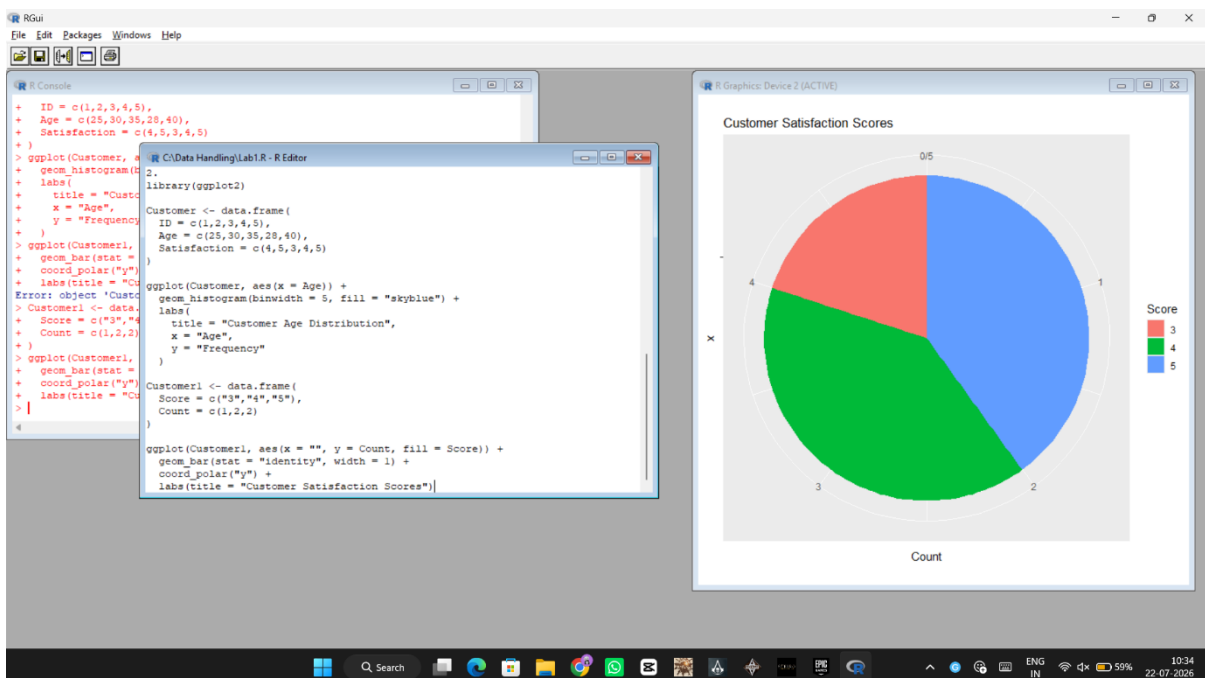


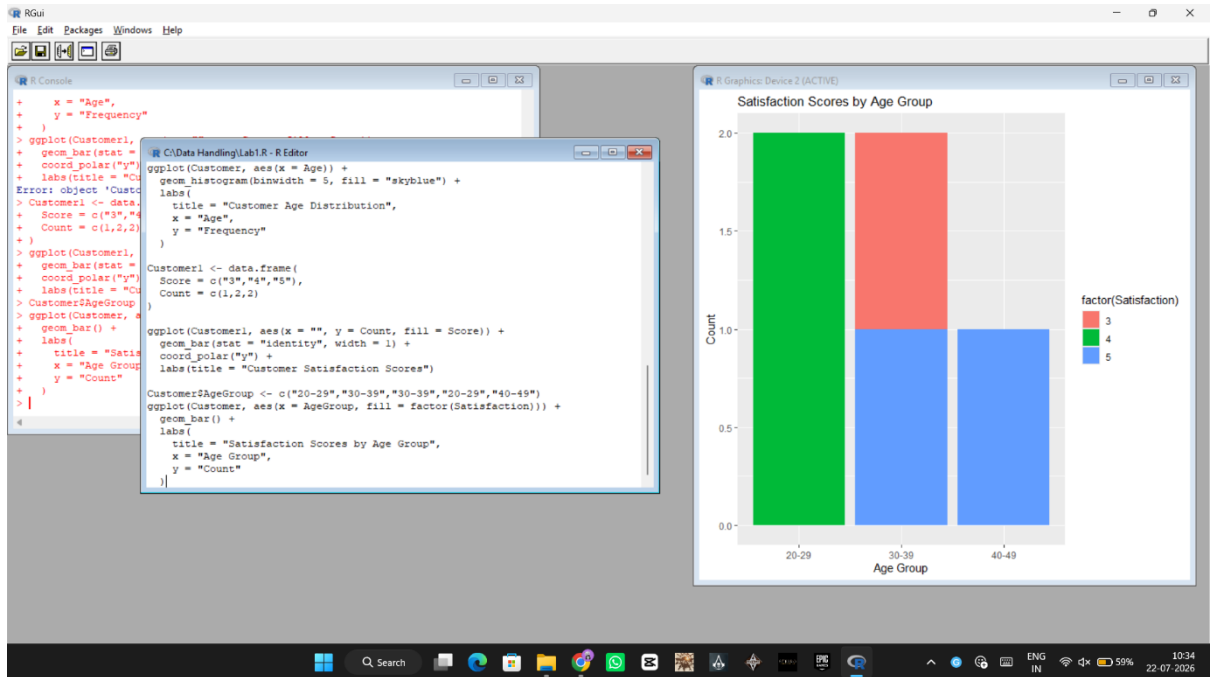
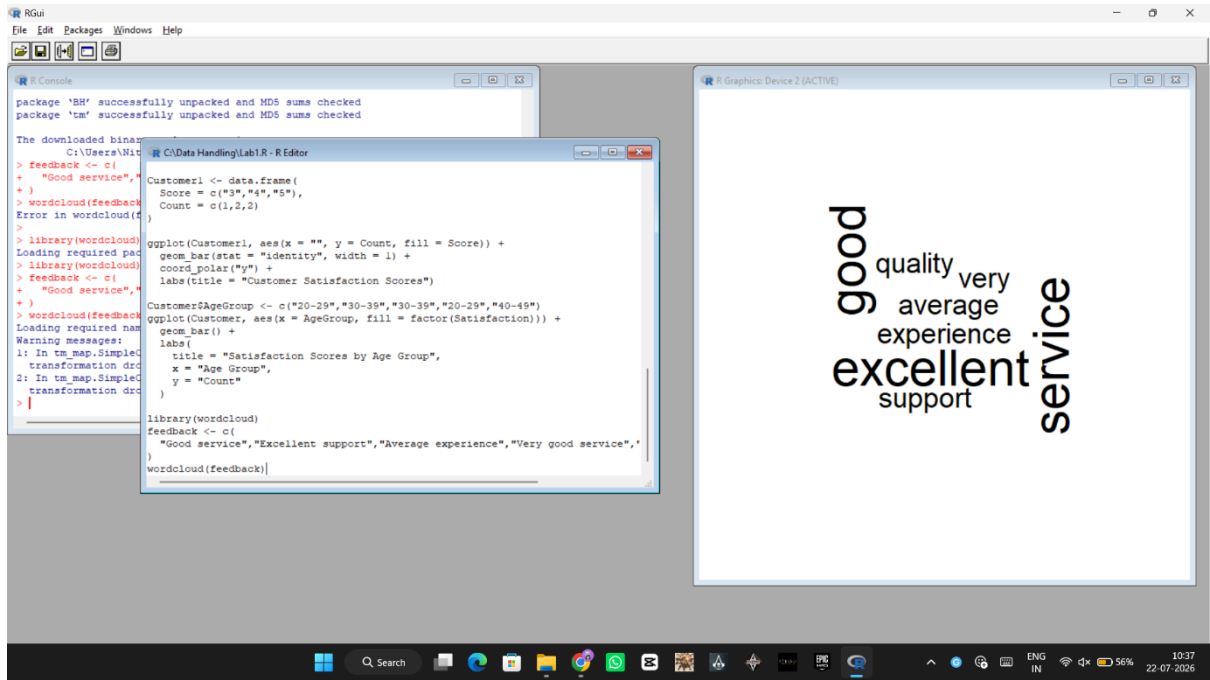


Set 2 : Customer Feedback Analysis

Dataset: Customer Satisfaction

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.
3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.
4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

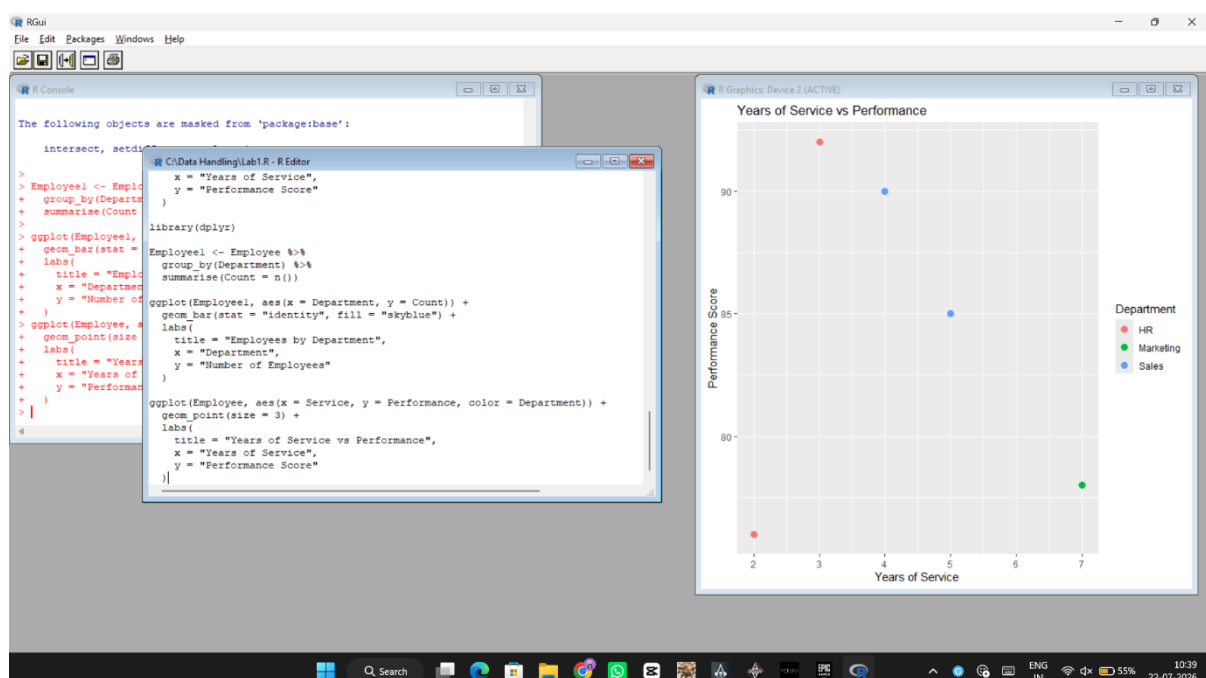
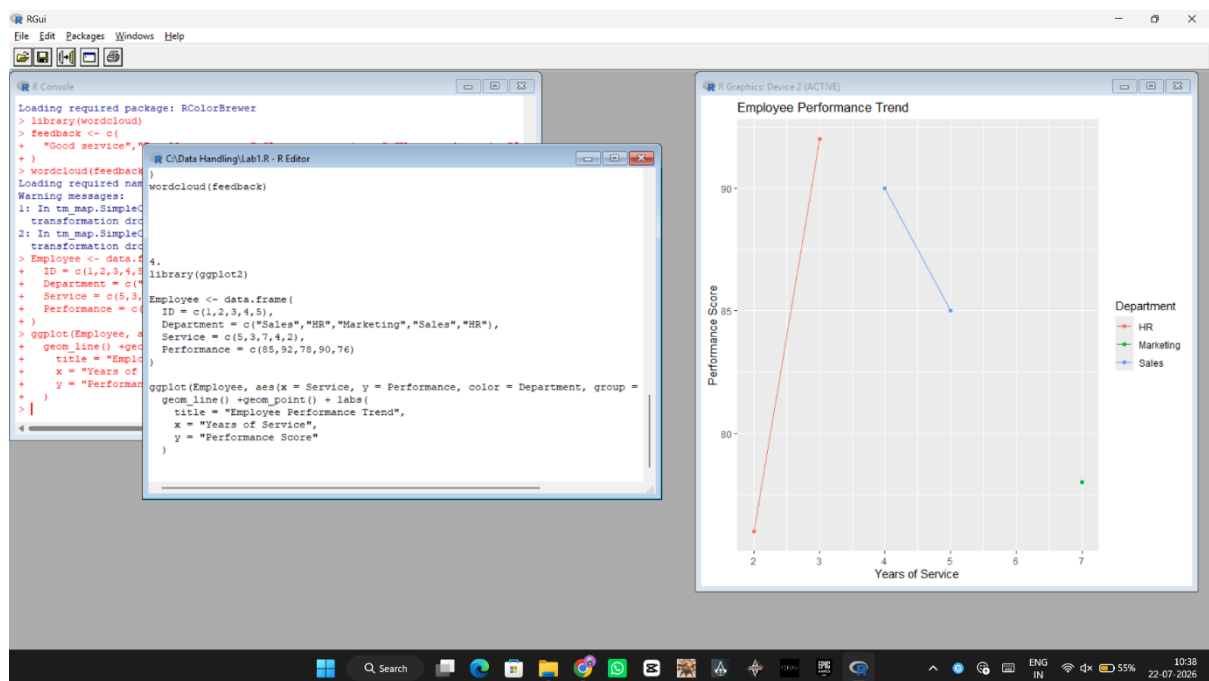


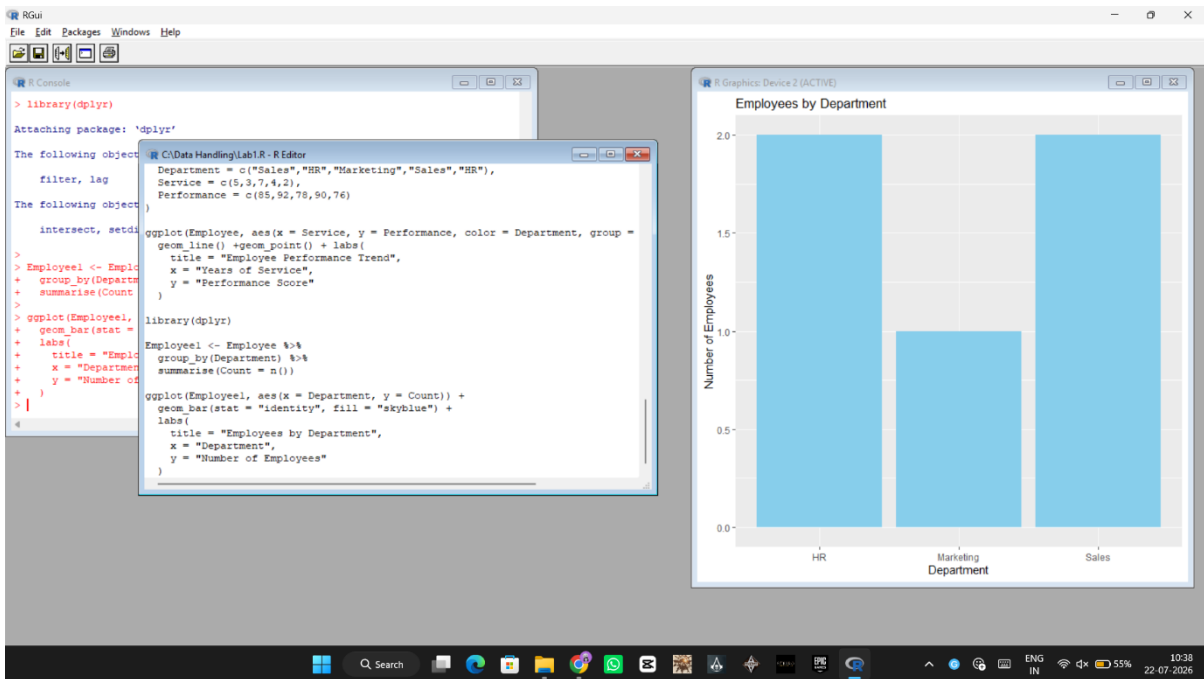


Set 3: Employee Performance Evaluation

Dataset: Employee Performance

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.
4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

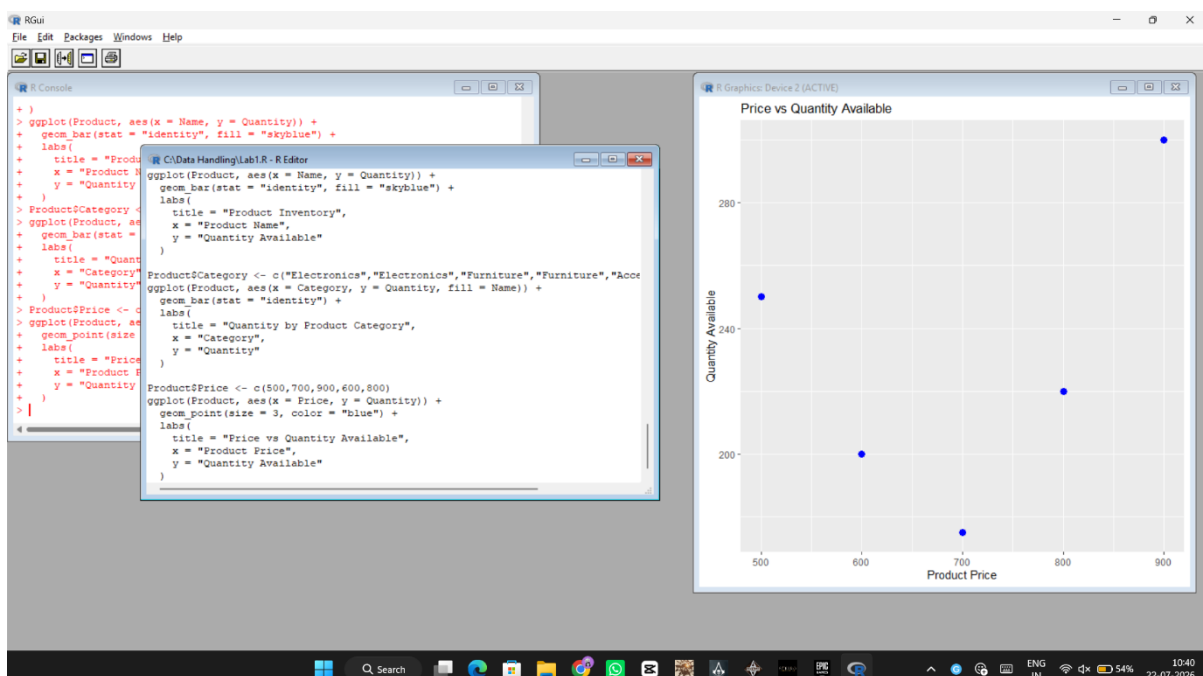


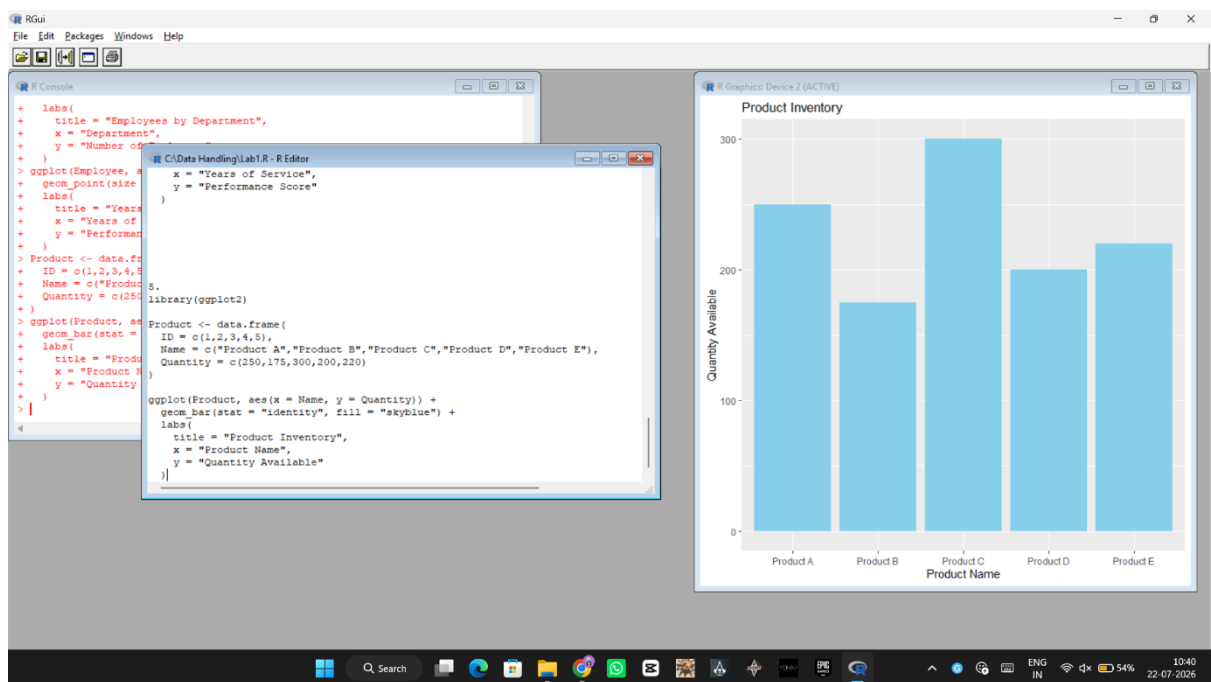
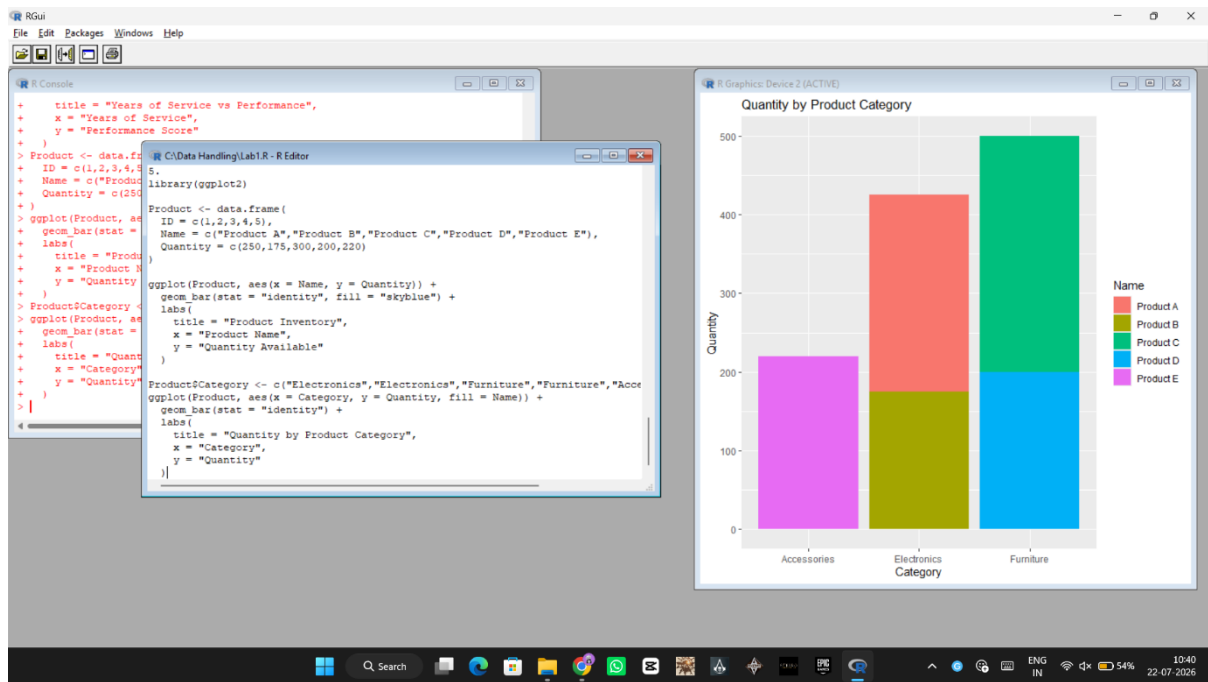


Set 4: Product Inventory Management

Dataset: Product Inventory

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.
4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.





Set 5: Website Analytics

Dataset: Website Traffic

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.
4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

